

Sneha Mishra

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LinkedIn | GitHub | LeetCode

SUMMARY

Engineering undergraduate with strong foundations in data structures, algorithms, and object-oriented programming. Experienced in designing and building scalable full-stack systems, translating real-world requirements into reliable software solutions, and collaborating with cross-functional stakeholders. Passionate about learning new technologies and building impactful, production-quality software.

EDUCATION

Indian Institute of Engineering Science and Technology, Shibpur Bachelor of Technology (Pre-final year) CGPA: 8.11/10.00	Howrah, India Aug 2023 – Present
Indian Institute of Technology Madras Bachelor of Data Science – Foundation Level CGPA: 7.71/10.00	India May 2024 – May 2025

TECHNICAL SKILLS

- **Programming Languages:** Java, C++, Python, JavaScript, TypeScript
- **Core CS:** Data Structures & Algorithms, Object-Oriented Programming, Operating Systems (basics), Computer Networks, Database Fundamentals
- **Web & Systems:** RESTful APIs, Authentication & Authorization (JWT), MVC Architecture
- **Databases:** MongoDB, MySQL
- **Frontend:** React.js, Next.js
- **Tools:** Git, GitHub

EXPERIENCE

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| Tata Steel – Research/Project Intern | Present |
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- Working under the guidance of **Alok Kumar, Head of TSCR Department**, on a project focused on optimization of heat flux to reduce crack breakout in TSCR (Thin Slab Casting and Rolling).
 - Utilizing **Python** for data analysis and heat transfer evaluation to identify factors affecting crack breakout, contributing to improved casting efficiency, reduced surface defects, and enhanced product quality.

Projects

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| Hostel Complaint & Application Management System | Node.js, Express.js, MongoDB |
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- Designed and developed a full-stack system to manage hostel complaints and administrative applications.
 - Implemented role-based access control for students and administrators using secure authentication.
 - Built a complete complaint lifecycle workflow with real-time status tracking (Pending, In-Progress, Resolved).
 - Developed admin features to review complaints, view uploaded images, add remarks, and update resolution status.
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| Credit Card Fraud Detection Using Machine Learning | Python, Scikit-learn, Pandas, NumPy |
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- Developed a machine learning model to detect fraudulent credit card transactions using Logistic Regression.
 - Performed data preprocessing, exploratory data analysis, and handled class imbalance through undersampling.
 - Trained and evaluated the model on a balanced dataset of 37,721 transactions with 30 features.
 - Achieved 97.48% training accuracy and 98.32% testing accuracy using Scikit-learn.
 - Built a prediction pipeline to classify transactions as legitimate or fraudulent transactions.

Positions of Responsibility

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| Team Member, EDC – IEST Shibpur | Aug 2024 – Present |
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- Planned onboarding and events for 100+ first-year students and coordinated workshops and mentorship programs.
- Organized speaker sessions and startup awareness events to introduce entrepreneurship opportunities on campus.
 - Collaborated with team members to increase event participation and engagement across departments.
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| Braking Lead, BAJA SAE Team – IEST Shibpur | May 2025 – Present |
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- Led brake system design calculations and collaborated on testing and optimization to meet safety and performance standards.
- Performed braking force and torque calculations to ensure compliance with BAJA SAE safety requirements.
 - Participated in testing, troubleshooting, and iterative improvements during vehicle trials.

Achievements

- **Flipkart GRID 7.0:** Reached National Semi-Finals among thousands of participants in Flipkart's flagship engineering competition.
- **Smart India Hackathon 2025:** Selected as college-level qualifier and developed a working solution under time-bound hackathon conditions.
- **Reliance Foundation Scholar:** Awarded a national merit-based scholarship.